MITCHELL GONDWE

BEng Computer and Electronic Engineering Student

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PROFESSIONAL SUMMARY

Motivated Computer and Electronic Engineering student with hands-on experience in Siemens NX, system integration, and technical documentation, gained through both academic and industry roles. Skilled at explaining technical concepts clearly and supporting others through complex problems, with experience assisting students and end-users in both engineering and programming contexts. Eager to bring this technical background and customer-focused approach to a Digital Industries customer services role.

SKILLS

Technical Proficiency: C, C++, Java, circuit design

Team Collaboration: Works effectively in cross-functional project teams

Adaptability: Quick to learn and apply new technologies and working environments

Community Engagement: Active volunteer in engineering outreach programmes

WORK EXPERIENCE

Vacation Work Trainee

Otto Wireless Solutions (Pty) Ltd, Randburg | June 2026 – July 2026

- Completed a five-week vacation work training programme covering IoT, LoRaWAN, RF and wireless technologies, cloud connectivity, and industrial networking.
- Gained hands-on experience in product development, system integration, and the creation of technical documentation.
- Deepened understanding of practical applications for wireless communication and IoT technologies.

Student Assistant

School of Computer Science and Information Systems, NWU | First Semester, 2024 – 2026

- Graded tests and assignments, invigilated examinations, and captured and uploaded marks for a C programming module, ensuring accurate assessment of student performance.
- Supported students with programming-related queries, reinforcing their understanding of key programming concepts through targeted assistance.

Temporary Assistant

Mechanical Engineering Department, NWU | First Semester, 2024 – 2026

- Marked Engineering Graphics assignments using Siemens NX, ensuring accuracy and adherence to technical drawing standards.
- Evaluated students' CAD models and technical drawings, providing constructive feedback to enhance their understanding of engineering design principles.
- Supported academic and administrative functions within the Mechanical Engineering Department, contributing to overall departmental efficiency.

EDUCATION

BEng Computer and Electronic Engineering

North-West University |02/2022 – 12/2026

Latest results: Databases and Web Programming 77% | Telecommunication Systems 70% | Control Theory II 69%|Embedded operating systems 61% |Signal theory 64% |Engineering management 60%

National Senior Certificate (Bachelor's Pass)

Victory House Private School |01/2016 – 12/2020

Matric results: Accounting 89% | Life Orientation 85% | Life Sciences 80% | English 74% | Mathematics 70% | Afrikaans 70% | Physics 70%

FINAL YEAR PROJECT

Android-Based Condition Monitoring for Induction Motors
BEng Mini-Dissertation, North-West University | 2026

- Designing a standalone Android application performing on-device, real-time vibration-based condition monitoring of induction motors, removing the need for internet connectivity or backend infrastructure, addressing a key limitation in remote agricultural and industrial environments.
- Implement a JPCCED-HI statistical health-index model, extracting 18 time- and frequency-domain features and applying Principal Component Analysis to classify motor condition and diagnose one of seven failure modes.
- Build a hybrid Kotlin + C++ (NDK) architecture, using C++ for computationally intensive signal processing (FFT, Hann windowing, feature extraction) to meet strict real-time processing requirements, achieving 290–665ms execution on mid-range hardware.

LANGUAGES

English: Fluent

Afrikaans: Intermediate

EXTRACURRICULAR ACTIVITIES

Facilitate workshops at Engenius Kids, a programme introducing engineering concepts to Grade 9 learners in township schools through engaging, hands-on projects, with the aim of encouraging continued study of Pure Mathematics and Physical Science.

REFERENCES

Dr Rodney Sebopelo
Senior Lecturer, North-West University

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Miyelani Kubayi
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